



STANDARD OPERATING PROCEDURE	MultiLink.Trial.SOP.10 v.1
Title: Patient education for self-management of multimorbidity (Intervention arm only)	
Effective date: 7 AUG. 2025	Next Review Due: 7 AUG. 2026

	Name and position
Author	Jennifer Ljungqvist, post-doctoral researcher
Reviewer	Eve Worrall, Co-PI
Reviewer	Martha Oshoseny, Social Scientist
Reviewer	Alice Rutta, Study Coordinator
Reviewer	Annamary Arcard Kajuna, Study Coordinator
Reviewer	Ndaona Mitole, Study Coordinator
Reviewer	Ben Morton, CI
Approver	Felix Limbani, Co-PI

REVISION HISTORY

Ver. No.	Revised by (Name & Position)	Effective Date	Details of changes
0.2	Jennifer Ljungqvist	17/07/2025	Minor additions to 4.2.1 and edits to appendices 1-3
0.3	Jennifer Ljungqvist	29/07/2025	Minor edits to align SOP with CRF questions and feedback from FL, including merging appendices 2 and 3.
0.4	Jennifer Ljungqvist	04/08/2025	Edited in response to EW review comments
1	Jennifer Ljungqvist	07/08/2025	Replaced 'treating clinicians' with 'research nurse'. As per BM request

1 BACKGROUND

This Standard Operating Procedure (SOP) has been developed to streamline the process of delivering patient education for self-care and self-management of multimorbidity for participants enrolled in **intervention sites** in the Multilink Trial.

2 PURPOSE

- 2.1 To outline the processes of providing study participant/patients and their caregivers with education for self-care and self-management of multimorbidity.
- 2.2 To describe the resources required for the efficient delivery of participant education for self-management of multimorbidity.
- 2.3 To detail approaches to evaluating adherence to SOP and effectiveness of the participant education intervention.

3 RESPONSIBILITY

Research nurses in intervention hospitals have the primary responsibility of delivering participant education and adhering to its protocol. However, each member of the study team is required to ensure adherence to participant education for self-management of multimorbidity protocols. The materials should also be shared with the treating clinicians and the participants caregivers to make them aware of the content covered in participant education sessions.

4 PROCEDURE

4.1 Materials

- Education **booklets** to be given to participants.
- Soft copy of health education materials with colourful illustrations to be shared with participants via for example WhatsApp in addition to hard copies, if they prefer it, and with stakeholders or partners as and when needed.
- **Flipbooks** for use by research study nurses during participant education sessions delivered in line with the patient visit schedule (inpatient, day 30, day 60, day 90) in Table 6 within the 'Multilink_Trial_Protocol_v5.0_02May2025'. Also use during training sessions with community care representatives and healthcare workers (HCWs).
- **Posters** that illustrate key lifestyle advice to be displayed in the hospital (with administrator and ward manager agreement) and can be referenced during one-to-one education sessions. It is highly encouraged that posters are also posted around the intervention hospitals in spaces where study participants are likely to see them (wards, waiting rooms, hallways etc.)
- Standardised **education checklist** (Appendix 2) for research nurses to ensure consistent delivery of the participant education sessions.
- The 'Participant Level Intervention' forms within the Case Report Form (CRF) in the trial database on REDCap, to assess the delivery of the participant education intervention and capture feedback from the participants on the extent to which they adopted disease self-management advice.

4.2 Methods

The flowchart Appendix 1 (**9.1**) in this document describes the process of participant education for self-management of multimorbidity in the intervention arm. The education checklists/topic guides in Appendix 2 (**9.2**) should guide the conversations help between research nurses and participants and/or their caregivers/guardians at the initial inpatient admission visit (**Table 1**) and the subsequent follow-up visits (**Table 2**) around day 30, 60

and 90 of their enrolment in the trial study. Below is a step-by-step procedure for delivering participant education.

4.2.1. Upon recruitment/enrolment into the study

- Upon recruitment (on day 2 or day of hospital discharge, whichever comes sooner) after the participant has received a diagnosis, the research nurse provides counselling to the participant by describing the conditions the participant has been diagnosed with/is being treated for.
- **Topics:** This information will include the definitions of the **conditions**, their **causes**, **signs**, and **symptoms**, and **what needs to happen now** that they have been diagnosed and will be detailed in printed booklets that the participants and their caregivers/guardians will be given to take home. The research nurse should focus on the information that is specific to the diseases that the participant has.
- **Location:** When providing this information research nurses will exercise utmost privacy and confidentiality and hold the conversation in a private space, if possible, cognisant of space limitations in these settings. Participant counselling will thus be provided on the bedside, in research clinics run by nurses, or in private consultation rooms.
- Posters will be displayed around the hospital in places that the participants and their caregivers/guardians are likely to see them. This could include waiting rooms, wards, or hallways. For areas outside the study room, research staff should agree on where to display the posters in agreement with the hospital management team.

4.2.2. Prior to discharge from the hospital

- Research nurses will conduct the participant education session on self-management of multimorbidity with the participant before the participant leaves the hospital, following the topics covered in the booklet and outlined in Appendix 2 (9.2).
- **Topics:** Research nurses will recap the disease-specific information provided during the counselling session (4.2.1) that took place at the time of recruitment and provide additional information on the **complications** that may occur due to living with multiple long-term conditions and especially if not taking medicine as advised.
 - o Research nurses will also walk the participants through a series of **lifestyle modification requirements** as provided for in the participant's education materials.
 - o The research nurse should focus on the information that is specific to the diseases that the participant has been diagnosed with. For the disease that the participant has *not* been diagnosed with, the research nurse should focus on the *preventive* measures that the participant can take to avoid getting this disease.
 - o Participants will be offered an opportunity to ask questions for clarification of additional information. Study research nurses are encouraged to consult with study clinical leads for help with questions that they cannot easily answer and then update the participant about what they have learned.
 - o Research nurses should provide a participant education **booklet** to the participant and/or to their caregiver and encourage them to read this in their own time. Where preferred, soft copies of the booklets can also be shared through WhatsApp with the participant and/or their caregiver.
 - o Research nurses should provide information on community care during these education sessions as well, as detailed in SOP 17 version 1.0 on community linkage.
 - o Towards the end of the initial education session at the first hospital visit, the research nurse will ask the participant a few questions to check whether they have understood the materials well, using the 'participant level intervention'

form in the CRF in REDCap. The study nurse should clarify any areas of confusion and make sure that they have answered the participant and/or caregiver's questions.

- Following each education session with a participant, research nurses will record details about the meeting in the REDCap CRF tool, using the 'Participant Level Interventions (Initial Visit)' form. The purpose of this data collection is to give the research nurses an opportunity to reflect on how well the education and community linkage session went, to understand any challenges experienced when delivering the participant education.
- **Delivery:** During the education sessions, research nurses should utilise the **flipbook** as a communication tool.
 - The education session will be conducted in the local national languages (Chichewa for Malawi and Kiswahili for Tanzania). Participant education can also be delivered in English if that is participant preference.
 - When providing education for self-management of multimorbidity, research nurses will encourage participants to be accompanied by a caregiver/guardian of their choice. It is important that caregivers/guardians are provided this information on best self-management practices for people with multimorbidity, considering their role in caring for the participant.
- **Location:** The education session should take place in a private setting, where possible, to exercise utmost privacy and confidentiality. This participant education will happen in the research clinics of the intervention hospitals.

4.2.3. Follow-up outpatient clinic appointment days 30, 60 and 90

- Participants will meet with the research nurse again at three subsequent visits to the intervention hospital on or close to day 30, 60 and 90 of their participation in the trial for a re-education session and discussion on self-management of multimorbidity and the use of community care structures/networks/organisations/services. These meetings should take place in a designated room at the hospital for examination and treatment according to the stipulated procedures.
- **Topics:** During these meetings, the research nurse will reengage the participant to discuss their conditions and lifestyle modification advice and address participant/caregiver-related queries.
 - Research nurses will also aim to understand from the participants any challenges associated with putting into practice any lifestyle modification advice, and whether any alterations were made according to the local context. In conversation with the participant, they will document these challenges and any lifestyle changes made in the REDCap CRF per participant (see instructions below).
 - The research nurse provides a recap to the participant about their conditions, what it means to have them, tips for self-management and potentially serious effects for not adhering to medication and modifying their lifestyle. When discussing potential serious effects from non-adherence, research nurses should carefully choose their words and be sensitive in their approach to not alarm the participants and/or their caregivers. The focus should be on helping participants and their caregivers feel empowered to manage the participants' conditions in a positive way that improves their life.
- During and following each education session, research nurses will record details about the meeting in the REDCap CRF tool per participant, using the 'Patient Level Interventions (Follow-up Visits)' form.
 - The first part of the form includes questions that the research nurse should fill out together with the participant and/or caregiver, capturing any lifestyle changes that the participant has reported making since their last hospital visit.

- The research nurse should fill in the second part of the form should be filled in after the participant and/or caregiver have left, providing feedback on if, when, how and who they conducted the re-education session with

5 HEALTH AND SAFETY

The health and safety section in the SOP focuses on ensuring the physical and psychological well-being of participants during the process. This includes:

1. Safety Protocols: Adherence to safety guidelines to protect participants and staff, especially in clinical settings. This includes the use of Personal Protective Equipment (PPE) as required.
2. Confidentiality: Strict maintenance of confidentiality for all participant information gathered during the process.
3. Discomfort mitigation: Procedures to minimize any stress or discomfort for participants during reminder and follow-up phone call or visits, including immediate response to any health concerns.
 - Research nurses should be sensitive in how they approach and talk to the participants and/or their caregivers, taking care to provide information in a conversational, friendly and positive way that makes the participant and/or their caregiver feel empowered. They should choose their language carefully to avoid scaring the participant and/or caregiver or inadvertently make them feel pressured to or insecure about making changes.
4. Safe Environment: Conducting the data collection in a private and comfortable setting.

6 ASSOCIATED PROCEDURES

- Clinical measurement
- Point-of-care test
- Community linkage intervention

7 REFERENCES

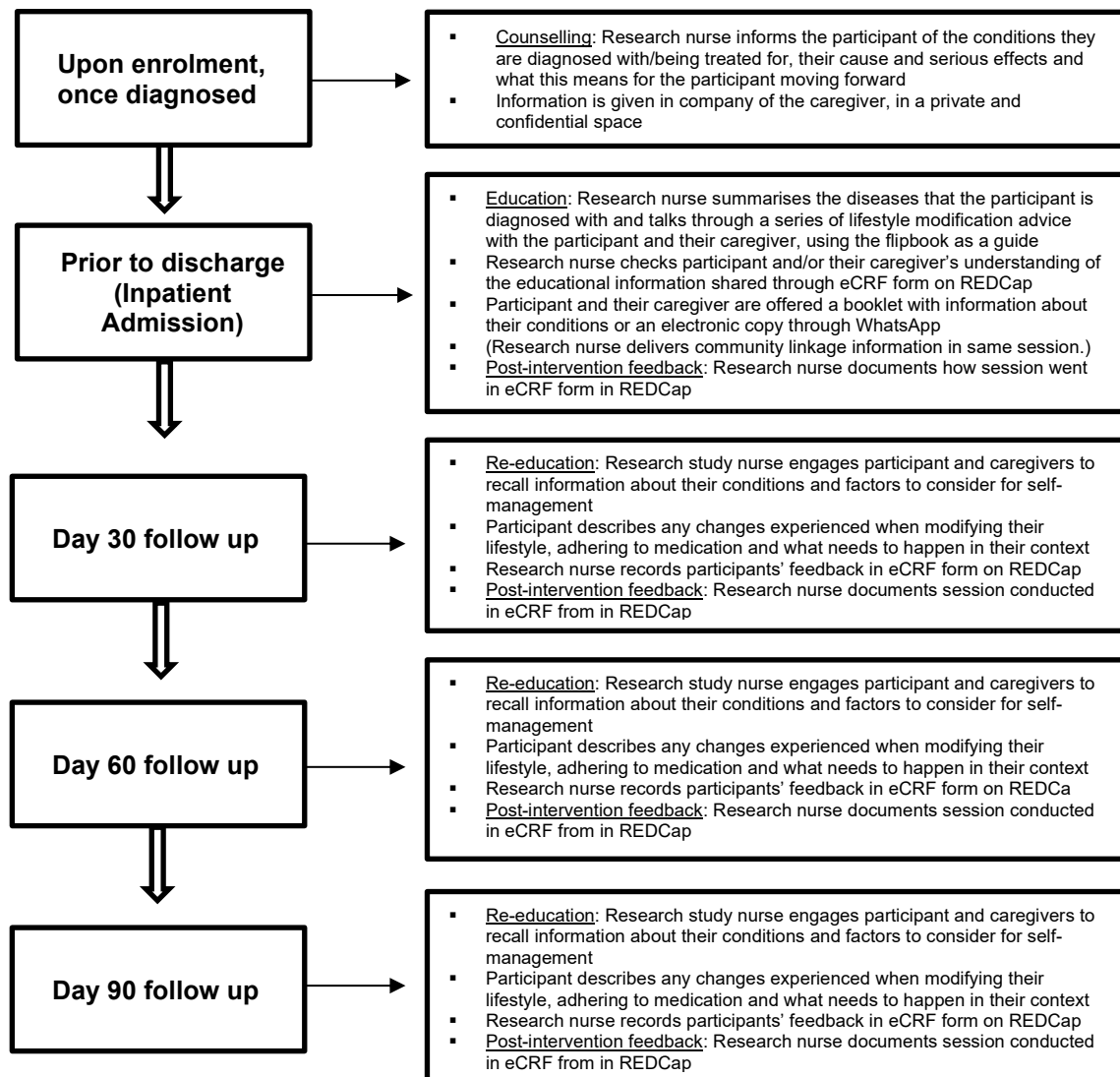
- Multilink_Trial_Protocol_v5.0_02May2025
- 17. MultiLink SOP_patient_community_linkage_v1.0

8 LIST OF ABBREVIATION

CI	Co-Investigator
CRF	Case Report Form
HCW	Healthcare Workers
PI	Principal Investigator
PPE	Personal Protective Equipment
SOP	Standard Operating Procedure

9 APPENDICES

9.1 Appendix 1: Intervention arm flow chart of delivering participant education for self-management



9.2 Appendix 2: Flow of contents for participant education for self-management of multimorbidity

Table 1: Topics Outline During First Hospital Visit (Inpatient Admission) and once recruited as a Study Participant in the Multilink Trial

Topic to Discuss	Study Nurse and Participant to Talk About
ONLY discuss the conditions/diseases that are relevant to the participant, based on their diagnosis. Any comments in relation to a disease that they have not been diagnosed with should be brief and focus on preventive measures.	
Diabetes	1. Definition of the disease 2. Causes of the disease 3. Signs, symptoms, and serious effects of the disease 4. What can I do? [Prevention and Self-management advice] 5. Challenges when taking medication
Hypertension	6. Definition of the disease 7. Causes of the disease 8. Signs, symptoms, and serious effects of the disease 9. What can I do? [Prevention and Self-management advice] 10. Challenges when taking medication
HIV/AIDS	11. Definition of the disease 12. Causes of the disease 13. Signs, symptoms, and serious effects of the disease 14. What can I do? [Prevention and Self-management advice] 15. Challenges when taking medication <i>Emergency Scenarios!</i> • Make sure that the participant understands at what point, with what symptoms, they should see medical advice.
Multimorbidity Clarify what specific challenges people with multimorbidity face	16. Definition of multimorbidity and specific challenges that people with multimorbidity face 17. When to seek help and from whom <i>Key messages!</i> • Adherence to taking <i>all relevant</i> medications • Prioritise advice from HCWs
Self-management <i>What a person with multiple chronic conditions can do to lead a healthier life at home</i>	18. Eat healthy food 19. Use less salt 20. Stop smoking/tobacco use 21. Use alcohol in moderation 22. Commit to taking medication regularly 23. Get physically active 24. Reduce stress
CRF questions in REDCap	25. Ask the participant-specific questions in the 'Participant Level Intervention' form in the CRF, to assess to what extent the participant understood and absorbed the information just discussed.

Questions and Answers	26. Listen. Answer the participant's questions, whether related to multimorbidity or not.
<i>Then discuss community link</i> Where should they go for routine clinical care in their local area	<i>Follow the procedures outlined in document '17. MultiLink SOP_patient_community_linkage'</i> • What organisations the participant can reach out to in the local area for physical health advice, mental wellbeing support and/or other lifestyle advice

Table 2: Topics Discussed At each Follow-up Visit on Day 30, Day 60, and/or Day 90

Topic to Discuss	Study Nurse and Participant to Talk About
Lifestyle changes since last visit	1. Summarise the participant education information that was covered during the first hospital visit (4.2.1) 2. Go through the participant-specific questions in the 'Participant Level Intervention' form in the CRF, while discussing if and how the participant may have changed their lifestyle. Or, if they have not made any changes to their routine, discuss what challenges they faced or the reasons why they did not engage in disease self-management activities. (Record reasons in REDCap eCRF form.)
Use of community care structures	3. Ask if they have used any community structures since their last visit and who these were, what services they accessed and their overall experience with these.

10 TRAINING RECORD FOR THIS SOP

The following have been trained in this SOP and have demonstrated adequate knowledge of its contents:

[illegible]